Project 1

a)

First I loaded the rat_pup data set to my R, and then set to work reproducing the tables and figures from chapter 3. Here is my table of weight statistics by splitting between treatment groups and sex:

```
Table:                Analysis Variable : weight
-----------------------------------------------------------------

|Treatment |Sex     | N Obs|  N| mean| std_dev| minimum| maximum|
|:---------|:------|-----:|--:|----:|-------:|-------:|-------:|
|High      |Female  |    32| 32| 5.85|    0.60|    4.48|    7.68|
|High      |Male    |    33| 33| 5.92|    0.69|    5.01|    7.70|
|Low       |Female  |    65| 65| 5.84|    0.45|    4.75|    7.73|
|Low       |Male    |    61| 61| 6.03|    0.38|    5.25|    7.13|
|Control   |Female  |    54| 54| 6.12|    0.69|    3.68|    7.57|
|Control   |Male    |    77| 77| 6.47|    0.75|    4.57|    8.33|
>
```

I couldn't find a graphic that grouped the treatments visually, so I opted for the table route, but all the information is still there. I'm not sure why they have an "N Obs" column and an "N" column, they seem to both imply the same thing, but I added it for consistency.

For figure 3.1, I had to do the same thing, but use a bwplot this time. Afterwards, I also added the mean as a diamond using a panel on top of the pre-existing bwplot. I reordered the treatment and sex factors to match the book, and got this. Visually, there is certainly a difference between the control and high / low groups, but as we'll come to find out later in model3.3, there is not enough evidence to show that high and low are statistically different, and we group the random factors together.

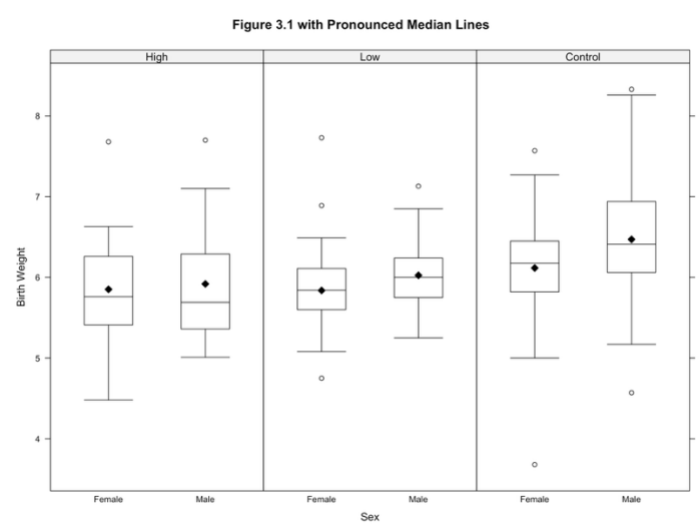
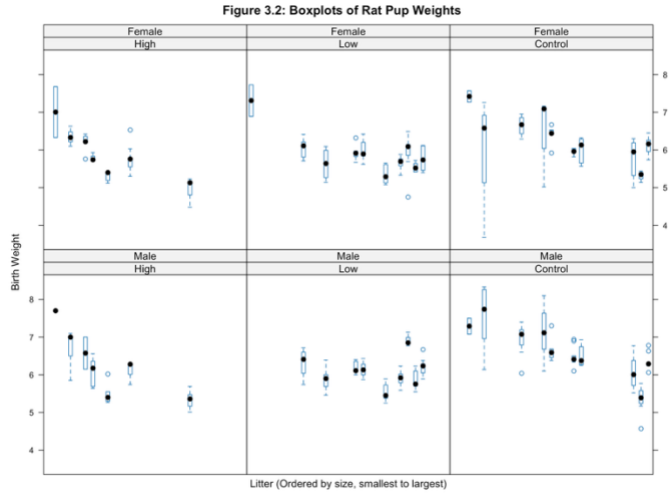


Figure 3.2 is another set of bwplots split by treatment and sex, but this time ordered by size of the rat litter. To do this, I ordered the data appropriately to match the book, and then drew weight ~ litter | treatment * sex to get the crossing.




This matches the book's production of this graph. As it mentions, weights decrease as a function of litter size, a trend we can see from these box plots.

b)

We start off with Hypothesis 3.1: that the random effects associated with litter-specific intercepts can be omitted from model 3.1. After creating model 3.1, we get this summary and Anova test. Our p-values are low for everything except the crossing of treatHi:sex1 and treatLow:sex1, given that both treat and sex are presupposed in the model (from the Anova test). Likely, we will remove this factor by the summary insignificance and the Anova test.

```
> print("------- Summary 3.1 ----------------------------------")
[1] "------- Summary 3.1 ----------------------------------"
> print(summary(model3.1.fit))
Linear mixed-effects model fit by REML
  Data: ratpup
       AIC      BIC    logLik
  419.1043 452.8775 -200.5522

Random effects:
 Formula: ~1 | litter
        (Intercept) Residual
StdDev:   0.3106722 0.404337

Fixed effects:  weight ~ treatment + sex1 + litsize + treatment:sex1
                      Value  Std.Error  DF   t-value p-value
(Intercept)        8.323340 0.27333009 292 30.451605  0.0000
treatmentHigh     -0.906057 0.19154238  23 -4.730320  0.0001
treatmentLow      -0.467040 0.15818328  23 -2.952521  0.0071
sex1              -0.411688 0.07315410 292 -5.627679  0.0000
litsize           -0.128382 0.01875336  23 -6.845819  0.0000
treatmentHigh:sex1  0.107023 0.13176318 292  0.812239  0.4173
treatmentLow:sex1   0.083866 0.10568189 292  0.793568  0.4281
 Correlation:
                  (Intr) trtmnH trtmnL sex1   litsiz trtH:1
treatmentHigh     -0.562
treatmentLow      -0.297  0.404
sex1              -0.111  0.158  0.191
litsize           -0.916  0.363  0.022  0.001
treatmentHigh:sex1  0.043 -0.323 -0.106 -0.555  0.020
treatmentLow:sex1   0.044 -0.096 -0.320 -0.692  0.036  0.385

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-7.47250744 -0.50014749  0.02911668  0.57348178  3.00962055

Number of Observations: 322
Number of Groups: 27
> print("---------------------------------- Anova 3.1 ----------------------------------
[1] "---------------------------------- Anova 3.1 ----------------------------------"
> print(anova(model3.1.fit))
               numDF denDF  F-value p-value
(Intercept)        1   292 9093.772  <.0001
treatment          2    23    5.082  0.0149
sex1               1   292   52.602  <.0001
litsize            1    23   47.374  <.0001
treatment:sex1     2   292    0.466  0.6282
>
```

However, we need to do a likelihood ratio test to confirm this against another model. After creating model 3.1a with no residual error variance among treatment groups, we get this anova summary:

```
[1] "----------------------------------- Anova 3.1 vs. 3.1A -----------------
> # Test Hypothesis 3.1.
> print(anova(model3.1.fit, model3.1a.fit))
              Model df      AIC      BIC    logLik   Test  L.Ratio p-value
model3.1.fit      1  9 419.1043 452.8775 -200.5522
model3.1a.fit     2  8 506.5099 536.5305 -245.2550 1 vs 2 89.40562  <.0001
>
```

Since our p-value is less than any strong significance level, we reject the null hypothesis that our residual error variances are all equal to 0. We should definitely keep different residual error variances among treatment groups, meaning treatment variance should be considered a random factor (at least using a random intercept).

```
[1] "------- Summary 3.2A ----------------------------------"
> print(summary(model3.2a.fit))
Linear mixed-effects model fit by REML
  Data: ratpup
       AIC     BIC    logLik
  381.8847 423.163 -179.9423

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.3134846 0.5147948

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | treatment
 Parameter estimates:
  Control       Low      High
1.0000000 0.5649830 0.6394383
Fixed effects:  weight ~ treatment + sex1 + litsize + treatment:sex1
                      Value  Std.Error  DF   t-value p-value
(Intercept)        8.345294 0.27464753 292 30.385468  0.0000
treatmentHigh     -0.903277 0.19215903  23 -4.700672  0.0001
treatmentLow      -0.466292 0.15908908  23 -2.931013  0.0075
sex1              -0.408131 0.09303486 292 -4.386865  0.0000
litsize           -0.130007 0.01848708  23 -7.032332  0.0000
treatmentHigh:sex1  0.094666 0.12919527 292  0.732737  0.4643
treatmentLow:sex1   0.076013 0.10811858 292  0.703053  0.4826
 Correlation:
                  (Intr) trtmnH trtmnL sex1   litsiz trtH:1
treatmentHigh     -0.598
treatmentLow      -0.362  0.466
sex1              -0.140  0.200  0.241
litsize           -0.905  0.376  0.056  0.000
treatmentHigh:sex1  0.087 -0.304 -0.173 -0.720  0.015
treatmentLow:sex1   0.100 -0.164 -0.303 -0.860  0.022  0.620

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.88670114 -0.52493419  0.02123518  0.57307286  2.56409983

Number of Observations: 322
Number of Groups: 27
>
```

Hypothesis 3.2: The variance of the residual errors is the same (homogeneous) for the three treatment groups (high, low, and control).

Model 3.2 helps us determine if each variance is different from each other. Our summary from 3.2 is this. Our parameter estimates for the variance function differ between control, Low, and High, which gives us insight to say we should consider

```
[1] "---------------------------------- Anova 3.1 vs. 3.2A -----------------
> # Test Hypothesis 3.2.
> print(anova(model3.1.fit, model3.2a.fit))
              Model df      AIC      BIC    logLik   Test  L.Ratio p-value
model3.1.fit      1  9 419.1043 452.8775 -200.5522
model3.2a.fit     2 11 381.8847 423.1630 -179.9423 1 vs 2 41.21964  <.0001
>
```

doing an Anova test against our previously chosen model 3.1. A p-value of less than .0001 tells us that the variance in residual errors is certainly non-zero.

Hypothesis 3.3: The residual error variances for the high and low treatment groups are equal

To test this, we create another model 3.2b, which groups the high and low treatment groups together. After fitting Model 3.2b and running an anova test compared to 3.2, our p-value is .2741. Since 3.2b is a nested model of 3.2a and 3.2b is simpler, we can select model 3.2b to continue with because we can combine the high and low treatment groups together, making it easier to explain and having less parameters, while still performing just as well as Model 3.2a.

```
__")
[1] "---------------------------------- Anova 3.2A vs. 3.2B ----------------
> print(anova(model3.2a.fit, model3.2b.fit))
              Model df      AIC      BIC    logLik   Test  L.Ratio p-value
model3.2a.fit     1 11 381.8847 423.1630 -179.9423
model3.2b.fit     2 10 381.0807 418.6065 -180.5404 1 vs 2 1.196053  0.2741
>
```

Hypothesis 3.4: The residual error variance for the combined high/low treatment group is equal to the residual error variance for the control group.

Hypothesis test 3.4 tells us whether we need to do all this splitting apart by treatment, or if we can group all the residual error variances together. Looking at the anova test, model 3.2b compared to model 3.1 has a p-value of <.0001, meaning it is better to fit the model with a difference in residual error variances compared to our equal variance model 3.1. Seeing the summary, we can conclude that there is a difference in the trtgrp parameters as far as a random intercept is concerned.

```
[1] "---------------------------------- Anova 3.1 vs. 3.2B ----------------
> print(anova(model3.1.fit, model3.2b.fit))
              Model df      AIC      BIC    logLik   Test  L.Ratio p-value
model3.1.fit      1  9 419.1043 452.8775 -200.5522
model3.2b.fit     2 10 381.0807 418.6065 -180.5404 1 vs 2 40.02358  <.0001
> print("------- Summary 3.2B ----------------------------------")
[1] "------- Summary 3.2B ----------------------------------"
> print(summary(model3.2b.fit))
Linear mixed-effects model fit by REML
  Data: ratpup
       AIC      BIC    logLik
  381.0807 418.6065 -180.5404

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.3145678 0.5147877

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | trtgrp
 Parameter estimates:
         1         2
1.0000000 0.5905488
Fixed effects:  weight ~ treatment + sex1 + litsize + treatment:sex1
                     Value  Std.Error  DF   t-value p-value
(Intercept)         8.350351 0.27567830 292 30.290200  0.0000
treatmentHigh      -0.901844 0.19140143  23 -4.711794  0.0001
treatmentLow       -0.466596 0.15999335  23 -2.916348  0.0078
sex1               -0.408195 0.09303538 292 -4.387530  0.0000
litsize            -0.130383 0.01856366  23 -7.023575  0.0000
treatmentHigh:sex1  0.092026 0.12461723 292  0.738473  0.4608
treatmentLow:sex1   0.076397 0.10939796 292  0.698337  0.4855
 Correlation:
                   (Intr) trtmnH trtmnL sex1   litsiz trtH:1
treatmentHigh      -0.605
treatmentLow       -0.358  0.466
sex1               -0.140  0.201  0.240
litsize            -0.906  0.382  0.054  0.000
treatmentHigh:sex1  0.092 -0.292 -0.178 -0.747  0.013
treatmentLow:sex1   0.097 -0.162 -0.306 -0.850  0.023  0.635

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.88725953 -0.52398488  0.01731336  0.56412336  2.55014720

Number of Observations: 322
Number of Groups: 27
>
```

Hypothesis 3.5: The fixed effects associated with the treatment by sex interaction are equal to zero in Model 3.2B.

To test hypothesis 3.5, we perform a simple F-test to see if the fixed effect of the crossing treatment & sex group is needed. Given our preceding add-1 effects from the anova test, we have a p-value of .7288 for adding treatment:sex, which is not significant enough to keep in our model.

```
[1] "---------------------------------- Anova 3.2B ----------------------
> print(anova(model3.2b.fit))
               numDF denDF  F-value p-value
(Intercept)        1   292 9027.743  <.0001
treatment          2    23    4.241  0.0271
sex1               1   292   61.568  <.0001
litsize            1    23   49.577  <.0001
treatment:sex1     2   292    0.317  0.7288
```

Moving forward, we will remove the interaction term for simplicity since it is not significant to include.

Finally,

```
[1] "---------------------------------- Anova 3.3 vs. 3.3A ------------------
> print(anova(model3.3.ml.fit, model3.3a.ml.fit))
                 Model df      AIC      BIC    logLik   Test  L.Ratio p-value
model3.3.ml.fit      1  8 353.7734 383.9698 -168.8867
model3.3a.ml.fit     2  6 368.3706 391.0179 -178.1853 1 vs 2 18.59723   1e-04
>
```

Hypothesis 3.6: The fixed effects associated with treatment are equal to zero in Model 3.3 (Primary research hypothesis, weight is affected by treatment group). We define model 3.3a to include the treatment as a fixed factor, which with a p-value of 1e-04, has a statistically significant effect on our model fit. We should keep the treatment group as a fixed factor in its effects on the output weight.

```
> print("------- Summary 3.3 ---------------------------------")
[1] "------- Summary 3.3 ---------------------------------"
> print(summary(model3.3.reml.fit))
Linear mixed-effects model fit by REML
  Data: ratpup
       AIC      BIC    logLik
  372.2784 402.3497 -178.1392

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.3146374 0.5144324

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | trtgrp
 Parameter estimates:
         1         2
1.0000000 0.5889108
Fixed effects:  weight ~ sex1 + litsize + treatment
                 Value  Std.Error  DF   t-value p-value
(Intercept)    8.327633 0.27406957 294 30.385106  0.0000
sex1          -0.343431 0.04204323 294 -8.168531  0.0000
litsize       -0.130681 0.01855194  23 -7.044036  0.0000
treatmentHigh -0.862268 0.18293359  23 -4.713556  0.0001
treatmentLow  -0.433663 0.15226167  23 -2.848140  0.0091
 Correlation:
              (Intr) sex1   litsiz trtmnH
sex1          -0.103
litsize       -0.913  0.043
treatmentHigh -0.610 -0.006  0.403
treatmentLow  -0.348 -0.035  0.064  0.464

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.97351508 -0.53695365  0.01508652  0.54234475  2.58286989

Number of Observations: 322
Number of Groups: 27
> print("-------  Anova 3.3 ---------------------------------")
[1] "-------  Anova 3.3 ---------------------------------"
> print(anova(model3.3.reml.fit))
            numDF denDF  F-value p-value
(Intercept)     1   294 9029.091  <.0001
sex1            1   294   63.596  <.0001
litsize         1    23   33.658  <.0001
treatment       2    23   11.387   4e-04
>
```

We output the final summary of model 3.3. our treatment group is included as a fixed factor along with sex and litsize, while the trtgrp is included as a random intercept factor. Each term included in the model significantly contributes to the model's fitting of weight based on the anova table on the bottom. This model is not considered for selection, only for research purposes

c)

From AIC, model 3.3a is the best model with the lowest AIC of 368.3706.

BIC also leads to the same conclusion, with a value of 391.0179.

Our fit is superior with model 3.3a, even though we chose 3.3 for the result since 3.3a was fit for research purposes.

d)

Here is figure 3.6, histograms for the conditional raw residuals of our fitted model. Both sets of residuals look approximately normal with slightly skewed tails on the left side.

Figure 3.7: The Q-Q plot of standard normal quantiles. Our line of best fit is overall good, save for a few outliers at the tails (especially the left tail)

Figure 3.8: Scatterplot of residuals to the predicted values. There is strong evidence here of nonconstant variance for the hi/low pooled group, however there was an outlier in the control group, pup_id = 66. It was poorly fitted, but did not have a significant effect on the output estimates.

Figure 3.9: The studentized residuals are approximately homogeneous across litters

Figure 3.6: Histograms of Conditional Raw Residuals

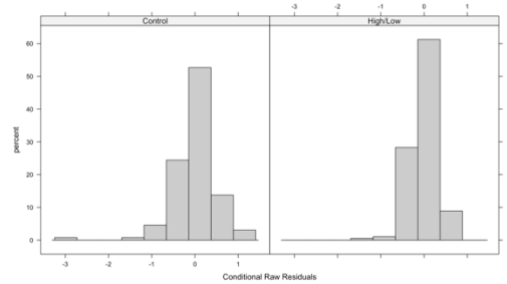



Figure 3.7: Normal Q-Q Plots of Conditional Raw Residuals

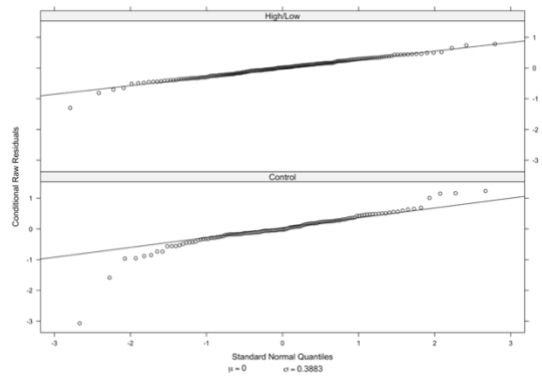
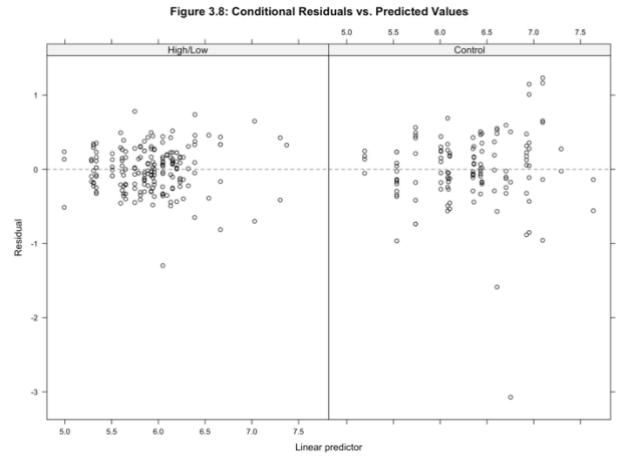



Figure 3.8: Conditional Residuals vs. Predicted Values



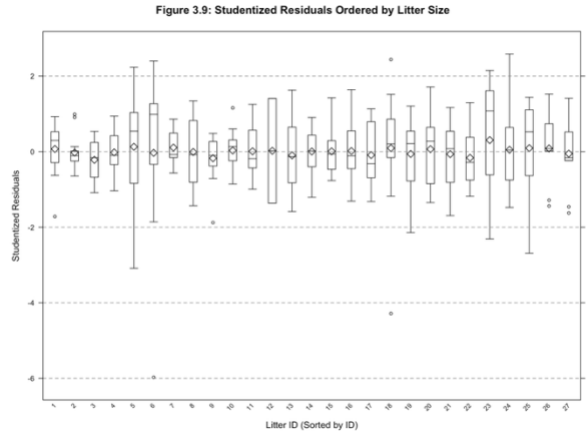


Figure 3.9: Studentized Residuals Ordered by Litter Size